sonam — -zsh — 80×24

```
zsh: command not found: y
sonam@Sonams-MacBook-Pro ~ % pip3 install --upgrade pip
Requirement already satisfied: pip in /Library/Frameworks/Python.framework/Versi
ons/3.11/lib/python3.11/site-packages (22.3.1)
Collecting pip
  Downloading pip-26.0.1-py3-none-any.whl (1.8 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.8/1.8 MB 10.3 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 22.3.1
    Uninstalling pip-22.3.1:
      Successfully uninstalled pip-22.3.1
Successfully installed pip-26.0.1
sonam@Sonams-MacBook-Pro ~ % pip3 install opencv-python
Collecting opencv-python
  Downloading opencv_python-4.13.0.92-cp37-abi3-macosx_13_0_arm64.whl.metadata (
19 kB)
Requirement already satisfied: numpy>=2 in /Library/Frameworks/Python.framework/
Versions/3.11/lib/python3.11/site-packages (from opencv-python) (2.4.3)
Downloading opencv_python-4.13.0.92-cp37-abi3-macosx_13_0_arm64.whl (46.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 46.2/46.2 MB 3.7 MB/s  0:00:12
Installing collected packages: opencv-python
Successfully installed opencv-python-4.13.0.92
sonam@Sonams-MacBook-Pro ~ %
```

python_basics Version Control

Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Assets
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
    - Task 1_4.py
    - Task 1_5.py
    - Task 1_6.py
    - Task 1_7.py
    - Task 1_8.py
    - Task 1_9.py
    - Task 1_10.py
    - Task 2_1.py
  - External Libraries
  - Scratches and Consoles

Task 1_1.py | Task 1_2.py | Task 1_3.py | Task 1_4.py | Task 1_5.py | Task 1_6.py | Task 1_7.py | Task 1_8.py | Task 1_9.py | Task 1_10.py | Task 2_1.py

```python
#your answer
#task2_1.py
import cv2
image = cv2.imread('Assets/Insects/1_Gammar 2021_1_2021_06_03-13-19-23-879.png')
print(f"Type of the image: {type(image)}")
print(f"Shape of the image array: {image.shape}")
```

Run Task 2_1

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 2_1.py
Type of the image: <class 'numpy.ndarray'>
Shape of the image array: (464, 464, 3)

Process finished with exit code 0
```

python_basics Version Control Current File Trial

Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Assets
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
    - Task 1_4.py
    - Task 1_5.py
    - Task 1_6.py
    - Task 1_7.py
    - Task 1_8.py
    - Task 1_9.py
    - Task 1_10.py
    - Task 2_1.py
    - Task 2_2.py
  - External Libraries
  - Scratches and Consoles

Task 1_1.py | Task 1_2.py | Task 1_3.py | Task 1_4.py | Task 1_5.py | Task 1_6.py | Task 1_7.py | Task 1_8.py | Task 1_9.py | Task 1_10.py | Task 2_1.py | Task 2_2.py

```python
#your answer
#task2_2.py
import cv2
image = cv2.imread('Assets/Insects/1_Gammar 2021_1_2021_06_03-13-19-23-879.png')
b, g, r = cv2.split(image)
cv2.imshow( winname: 'Blue Channel', b)
cv2.imshow( winname: 'Green Channel', g)
cv2.imshow( winname: 'Red Channel', r)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Run Task 2_2

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 2_2.py
```

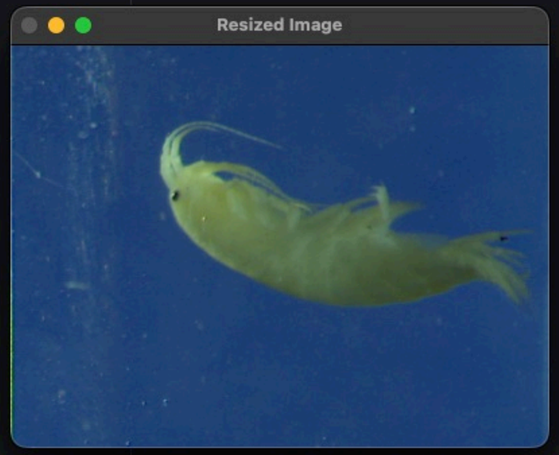
python_basics Version Control Current File Trial

Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Assets
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
    - Task 1_4.py
    - Task 1_5.py
    - Task 1_6.py
    - Task 1_7.py
    - Task 1_8.py
    - Task 1_9.py
    - Task 1_10.py
    - Task 2_1.py
    - Task 2_2.py
    - Task 2_3.py
  - External Libraries
  - Scratches and Consoles

Task 1_1.py | Task 1_2.py | Task 1_3.py | Task 1_4.py | Task 1_5.py | Task 1_6.py | Task 1_7.py | Task 1_8.py | Task 1_9.py | Task 1_10.py | Task 2_1.py | Task 2_2.py | Task 2_3.py

```python
#your answer
#task2_3.py
import cv2
image = cv2.imread('Assets/Insects/1_Gammar 2021_1_2021_06_03-13-19-23-879.png')
resized_img = cv2.resize(image, dsize: (400, 300))
cv2.imshow( winname: 'Resized Image', resized_img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



Run Task 2_2 Task 2_3

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 2_3.py
```

PythonProject > Week5LabActivity > Task 2_3.py 5:44 LF UTF-8 4 spaces Python 3.11 (PythonProject)

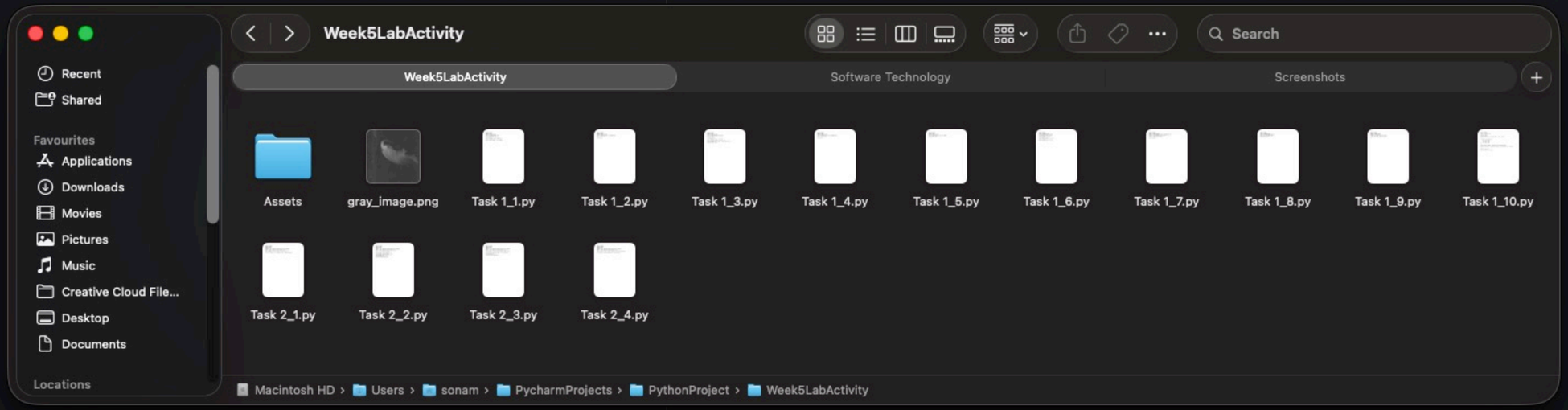
python_basics Version Control Current File

Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Assets
    - gray_image.png
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
    - Task 1_4.py
    - Task 1_5.py
    - Task 1_6.py
    - Task 1_7.py
    - Task 1_8.py
    - Task 1_9.py
    - Task 1_10.py
    - Task 2_1.py
    - Task 2_2.py
    - Task 2_3.py
    - Task 2_4.py
- External Libraries
- Scratches and Consoles

Task 1_1.py | Task 1_2.py | Task 1_3.py | Task 1_4.py | Task 1_5.py | Task 1_6.py | Task 1_7.py | Task 1_8.py | Task 1_9.py | Task 1_10.py | Task 2_1.py | Task 2_2.py | Task 2_3.py | Task 2_4.py

```python
#your answer
#task2_4.py
import cv2
image = cv2.imread('Assets/Insects/1_Gammar 2021_1_2021_06_03-13-19-23-879.png')
gray_image = cv2.cvtColor(image,cv2.COLOR_BGR2GRAY)
cv2.imwrite( filename: 'gray_image.png', gray_image)
```



Run Task 2_2 | Task 2_3 | Task 2_4

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 2_4.py

Process finished with exit code 0
```

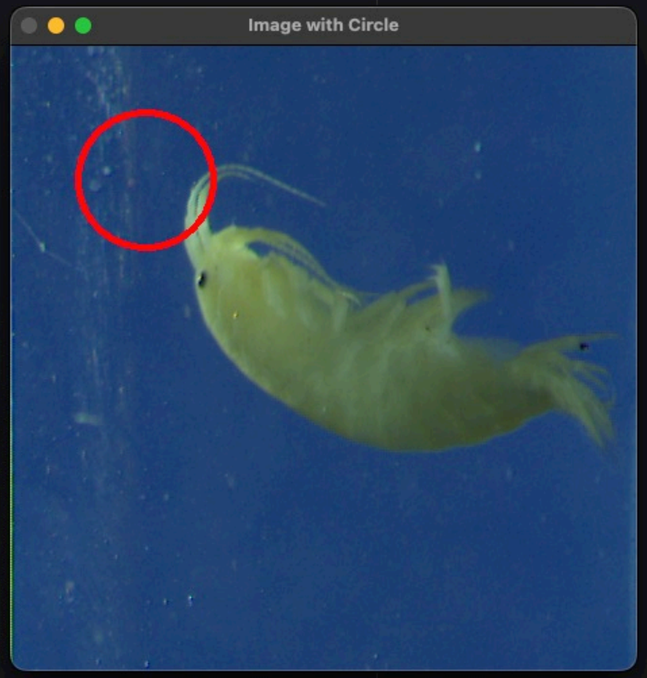
python_basics Version Control Current File Trial

Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Assets
    - gray_image.png
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
    - Task 1_4.py
    - Task 1_5.py
    - Task 1_6.py
    - Task 1_7.py
    - Task 1_8.py
    - Task 1_9.py
    - Task 1_10.py
    - Task 2_1.py
    - Task 2_2.py
    - Task 2_3.py
    - Task 2_4.py
    - Task 2_5.py
  - External Libraries
  - Scratches and Consoles

Task 1_1.py Task 1_2.py Task 1_3.py Task 1_4.py Task 1_5.py Task 1_6.py Task 1_7.py Task 1_8.py Task 1_9.py Task 1_10.py Task 2_1.py Task 2_2.py Task 2_3.py Task 2_4.py Task 2_5.py

```python
#your answer
#task2_5.py
import cv2
image = cv2.imread('Assets/Insects/1_Gammar 2021_1_2021_06_03-13-19-23-879.png')
image_copy = image.copy()
cv2.circle(image_copy, center: (100, 100), radius: 50, color: (0, 0,
255), thickness: 3) # BGR: Red
cv2.imshow( winname: 'Image with Circle', image_copy)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



Run Task 2_2 Task 2_3 Task 2_5

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 2_5.py
```

PythonProject > Week5LabActivity > Task 2_5.py 4:75 LF UTF-8 4 spaces Python 3.11 (PythonProject)

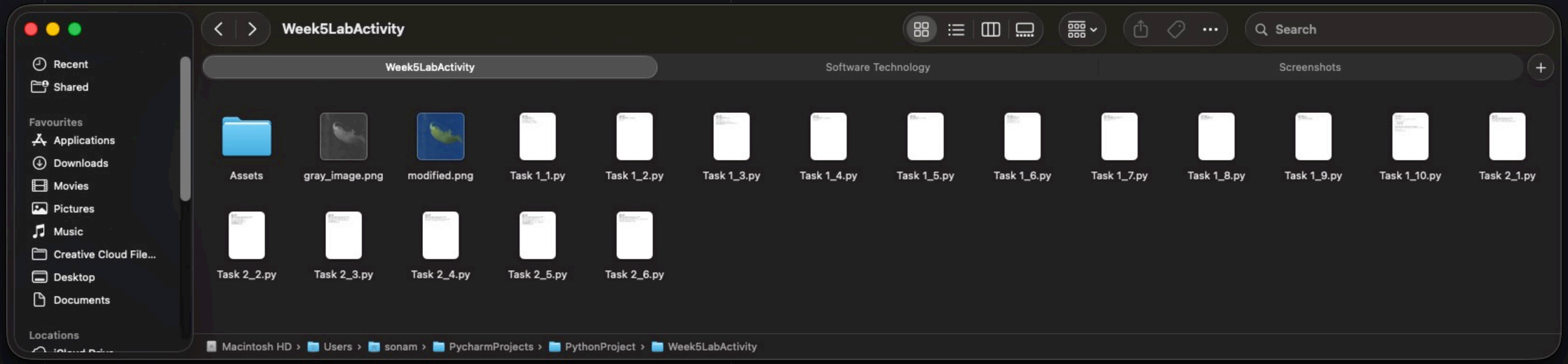
python_basics Version Control Current File

Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Assets
    - gray_image.png
    - modified.png
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
    - Task 1_4.py
    - Task 1_5.py
    - Task 1_6.py
    - Task 1_7.py
    - Task 1_8.py
    - Task 1_9.py
    - Task 1_10.py
    - Task 2_1.py
    - Task 2_2.py
    - Task 2_3.py
    - Task 2_4.py
    - Task 2_5.py
    - Task 2_6.py
  - External Libraries
  - Scratches and Consoles

Task 1_1.py | Task 1_2.py | Task 1_3.py | Task 1_4.py | Task 1_5.py | Task 1_6.py | Task 1_7.py | Task 1_8.py | Task 1_9.py | Task 1_10.py | Task 2_1.py | Task 2_2.py | Task 2_3.py | Task 2_4.py | Task 2_5.py | Task 2_6.py

```python
#your answer
#task2_6.py
import cv2
image = cv2.imread('Assets/Insects/1_Gammar 2021_1_2021_06_03-13-19-23-879.png')
print(f"Original pixel value at (50, 50): {image[50, 50]}")
image[50, 50] = [255, 255, 255]
cv2.imwrite( filename: 'modified.png', image)
```



Run: Task 2_2 | Task 2_3 | Task 2_5 | Task 2_6

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 2_6.py
Original pixel value at (50, 50): [120  65  29]

Process finished with exit code 0
```

python_basics Version Control Current File Trial

Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Assets
      - gray_image.png
      - modified.png
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
    - Task 1_4.py
    - Task 1_5.py
    - Task 1_6.py
    - Task 1_7.py
    - Task 1_8.py
    - Task 1_9.py
    - Task 1_10.py
    - Task 2_1.py
    - Task 2_2.py
    - Task 2_3.py
    - Task 2_4.py
    - Task 2_5.py
    - Task 2_6.py
    - Task 2_7.py
  - External Libraries
  - Scratches and Consoles

Task 1_2.py | Task 1_3.py | Task 1_4.py | Task 1_5.py | Task 1_6.py | Task 1_7.py | Task 1_8.py | Task 1_9.py | Task 1_10.py | Task 2_1.py | Task 2_2.py | Task 2_3.py | Task 2_4.py | Task 2_5.py | Task 2_6.py | Task 2_7.py

```python
#your answer
#task2_7.py
import cv2
import numpy as np
image = cv2.imread('Assets/Insects/1_Gammar 2021_1_2021_06_03-13-19-23-879.png')
inverted = 255 - image
combined = np.hstack((image, inverted))
cv2.imshow( winname: 'Original and Inverted', combined)
cv2.waitKey(0)
```

Run Task 2_2 Task 2_3 Task 2_5 Task 2_7

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 2_7.py
```

PythonProject > Week5LabActivity > Task 2_7.py 5:75 LF UTF-8 4 spaces Python 3.11 (PythonProject)

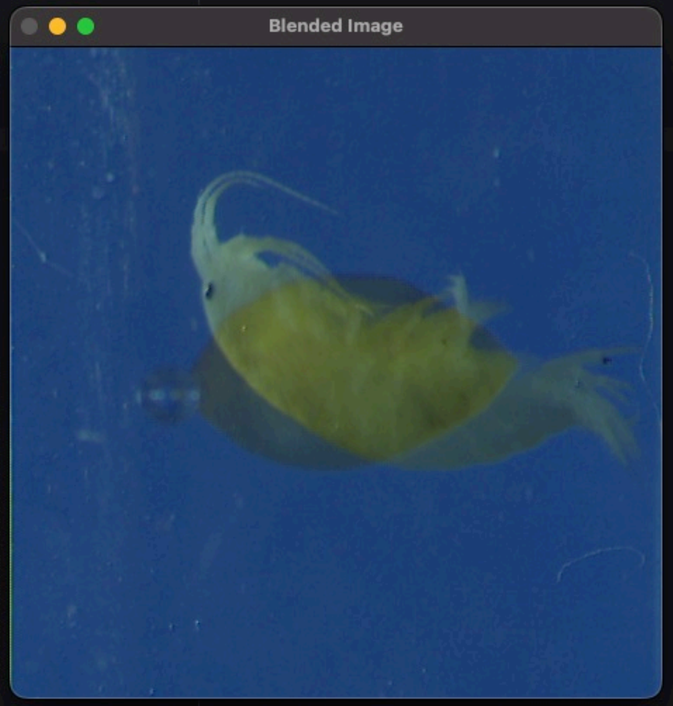
python_basics Version Control Current File

Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Assets
    - gray_image.png
    - modified.png
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
    - Task 1_4.py
    - Task 1_5.py
    - Task 1_6.py
    - Task 1_7.py
    - Task 1_8.py
    - Task 1_9.py
    - Task 1_10.py
    - Task 2_1.py
    - Task 2_2.py
    - Task 2_3.py
    - Task 2_4.py
    - Task 2_5.py
    - Task 2_6.py
    - Task 2_7.py
    - Task 2_8.py
  - External Libraries
  - Scratches and Consoles

Task 1_3.py Task 1_4.py Task 1_5.py Task 1_6.py Task 1_7.py Task 1_8.py Task 1_9.py Task 1_10.py Task 2_1.py Task 2_2.py Task 2_3.py Task 2_4.py Task 2_5.py Task 2_6.py Task 2_7.py Task 2_8.py

```python
#your answer
#task2_8.py
import cv2
img1 = cv2.imread('Assets/Insects/1_Gammar 2021_1_2021_06_03-13-19-23-879.png')
img2 = cv2.imread('Assets/Insects/2_Sphaer 2021_1_2021_06_03-11-22-54-852.png')
img2_resized = cv2.resize(img2, dsize: (img1.shape[1], img1.shape[0]))
blended = cv2.addWeighted(img1, alpha: 0.6, img2_resized, beta: 0.4, gamma: 0)
cv2.imshow( winname: 'Blended Image', blended)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



Run Task 2_2 Task 2_3 Task 2_5 Task 2_8

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 2_8.py
```

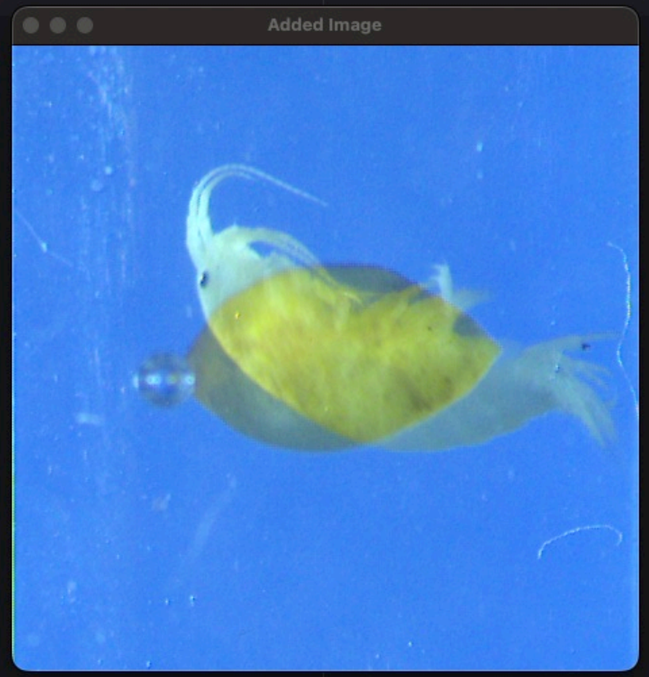
python_basics Version Control Current File Trial

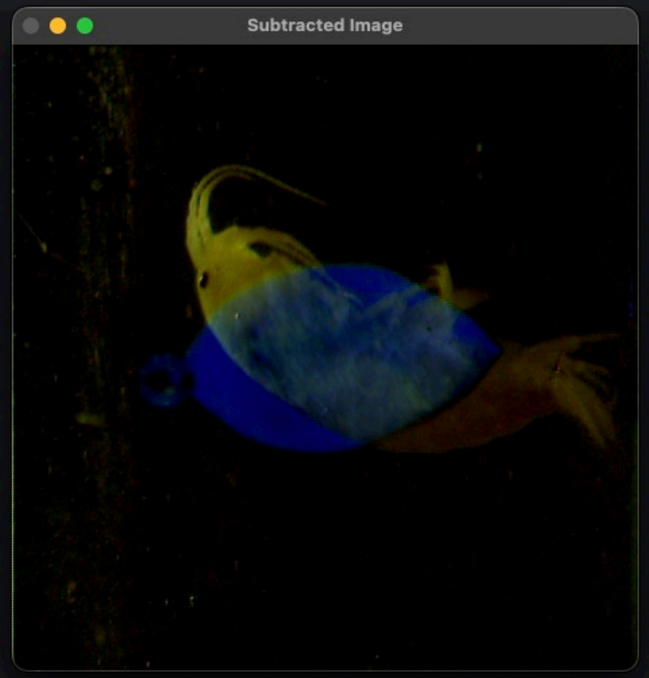
Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Assets
      - gray_image.png
      - modified.png
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
    - Task 1_4.py
    - Task 1_5.py
    - Task 1_6.py
    - Task 1_7.py
    - Task 1_8.py
    - Task 1_9.py
    - Task 1_10.py
    - Task 2_1.py
    - Task 2_2.py
    - Task 2_3.py
    - Task 2_4.py
    - Task 2_5.py
    - Task 2_6.py
    - Task 2_7.py
    - Task 2_8.py
    - Task 2_9.py
  - External Libraries
  - Scratches and Consoles

Task 1_4.py | Task 1_5.py | Task 1_6.py | Task 1_7.py | Task 1_8.py | Task 1_9.py | Task 1_10.py | Task 2_1.py | Task 2_2.py | Task 2_3.py | Task 2_4.py | Task 2_5.py | Task 2_6.py | Task 2_7.py | Task 2_8.py | Task 2_9.py

```python
#your answer
#task2_9.py
import cv2
img1 = cv2.imread('Assets/Insects/1_Gammar 2021_1_2021_06_03-13-19-23-879.png')
img2 = cv2.imread('Assets/Insects/2_Sphaer 2021_1_2021_06_03-11-22-54-852.png')
img2_resized = cv2.resize(img2, dsize: (img1.shape[1], img1.shape[0]))
added = cv2.add(img1, img2_resized)
subtracted = cv2.subtract(img1, img2_resized)
cv2.imshow( winname: 'Added Image', added)
cv2.imshow( winname: 'Subtracted Image', subtracted)
cv2.waitKey(0)
cv2.destroyAllWindows()
```





Run Task 2_2 | Task 2_3 | Task 2_5 | Task 2_8 | Task 2_9

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 2_9.py
```

PythonProject > Week5LabActivity > Task 2_9.py 5:74 LF UTF-8 4 spaces Python 3.11 (PythonProject)

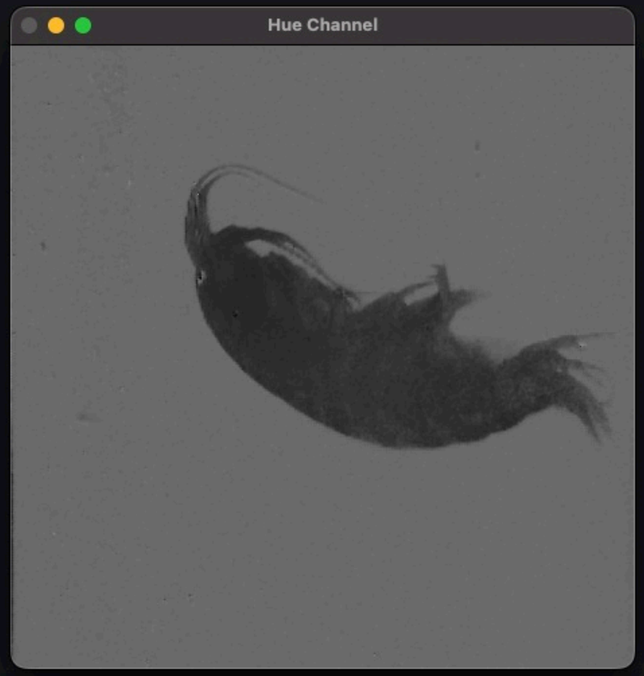
python_basics Version Control Current File Trial

Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Assets
      - gray_image.png
      - modified.png
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
    - Task 1_4.py
    - Task 1_5.py
    - Task 1_6.py
    - Task 1_7.py
    - Task 1_8.py
    - Task 1_9.py
    - Task 1_10.py
    - Task 2_1.py
    - Task 2_2.py
    - Task 2_3.py
    - Task 2_4.py
    - Task 2_5.py
    - Task 2_6.py
    - Task 2_7.py
    - Task 2_8.py
    - Task 2_9.py
    - Task 2_10.py
  - External Libraries
  - Scratches and Consoles

Task 1_5.py | Task 1_6.py | Task 1_7.py | Task 1_8.py | Task 1_9.py | Task 1_10.py | Task 2_1.py | Task 2_2.py | Task 2_3.py | Task 2_4.py | Task 2_5.py | Task 2_6.py | Task 2_7.py | Task 2_8.py | Task 2_9.py | Task 2_10.py

```python
#your answer
#task2_10.py
import cv2
image = cv2.imread('Assets/Insects/1_Gammar 2021_1_2021_06_03-13-19-23-879.png')
hsv_img = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)
h, s, v = cv2.split(hsv_img)
cv2.imshow( winname: 'Hue Channel', h)
cv2.imshow( winname: 'Saturation Channel', s)
cv2.imshow( winname: 'Value Channel', v)
cv2.waitKey(0)
```



Run: Task 2_2 | Task 2_3 | Task 2_5 | Task 2_8 | Task 2_10

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 2_10.py
```

PythonProject > Week5LabActivity > Task 2_10.py 4:75 LF UTF-8 4 spaces Python 3.11 (PythonProject)